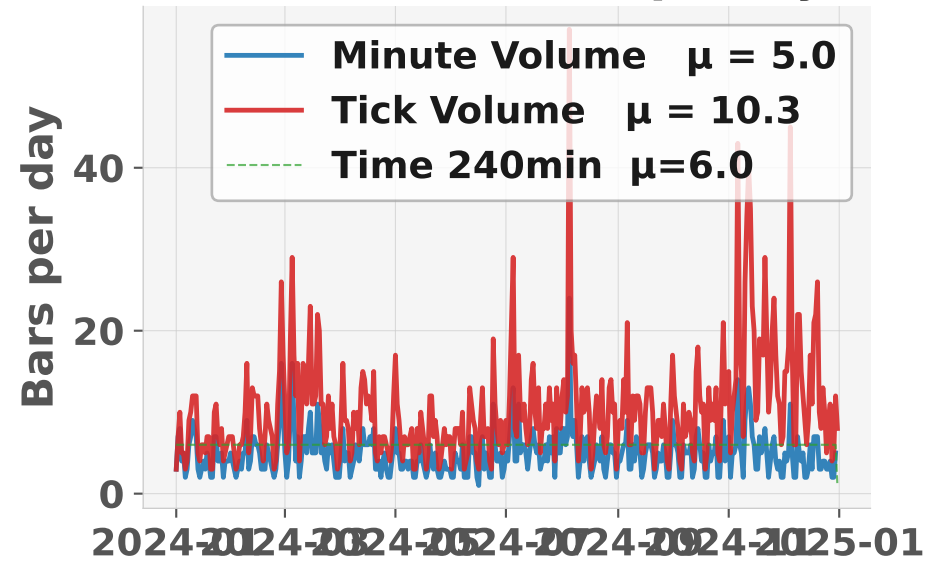
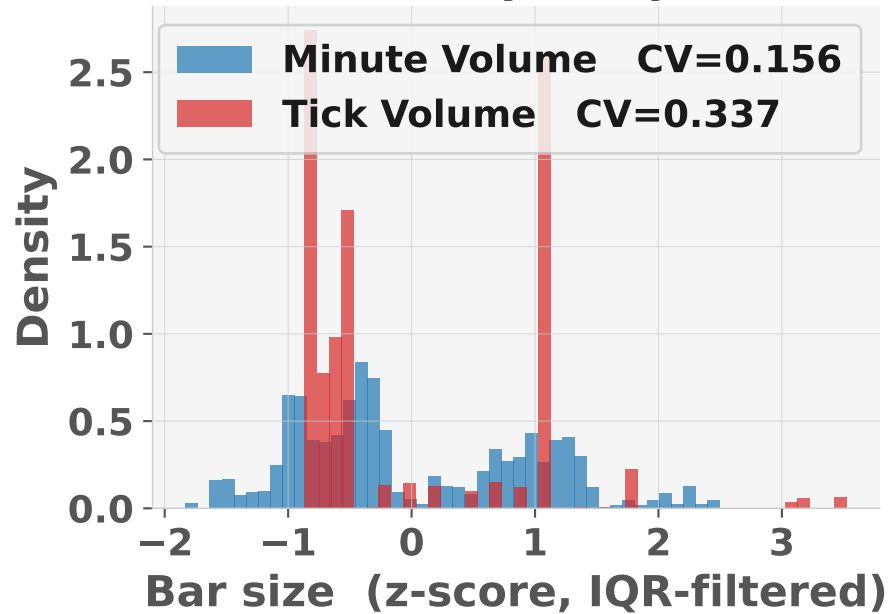
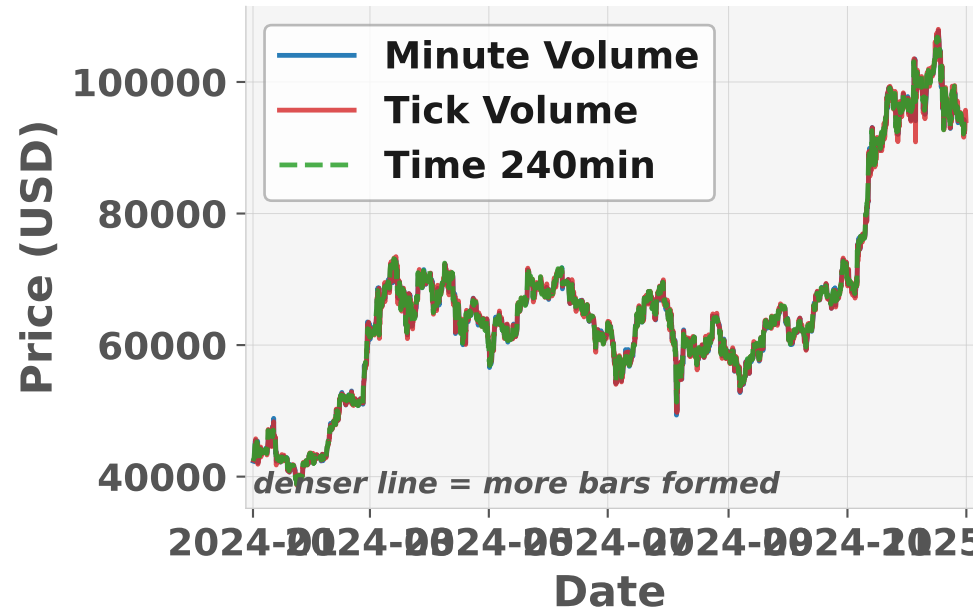
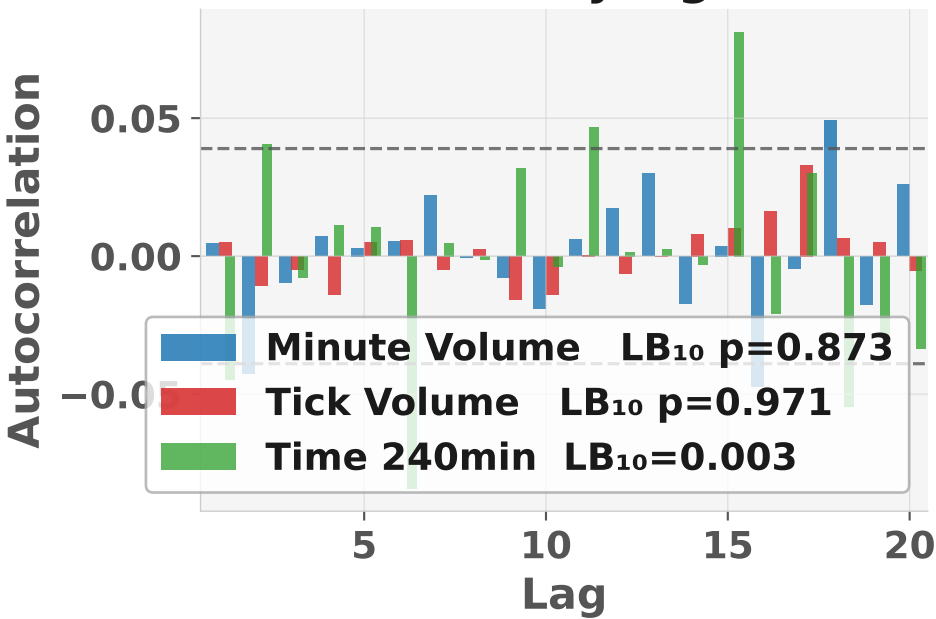
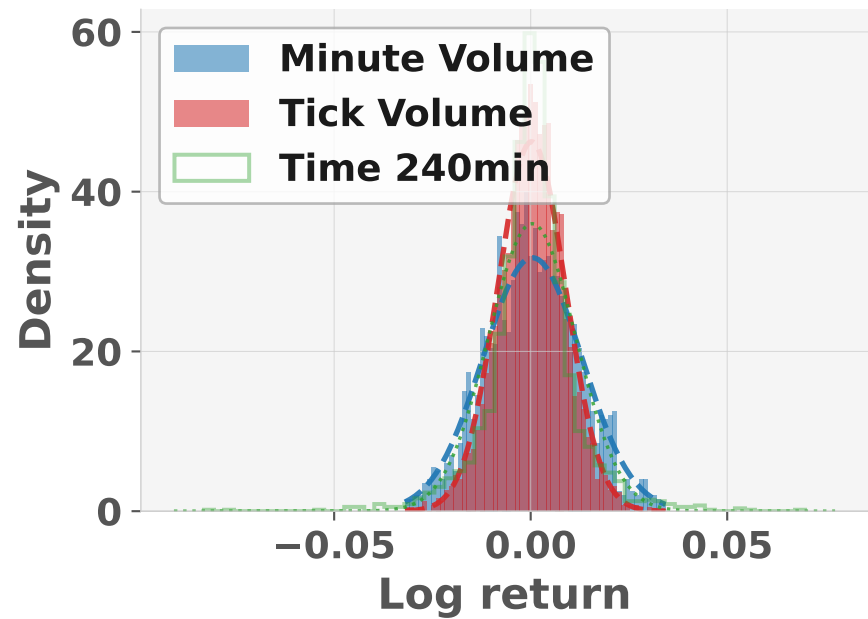


Information Bar Comparison: Minute Volume vs Tick Volume vs Time 240min (time bar baseline)

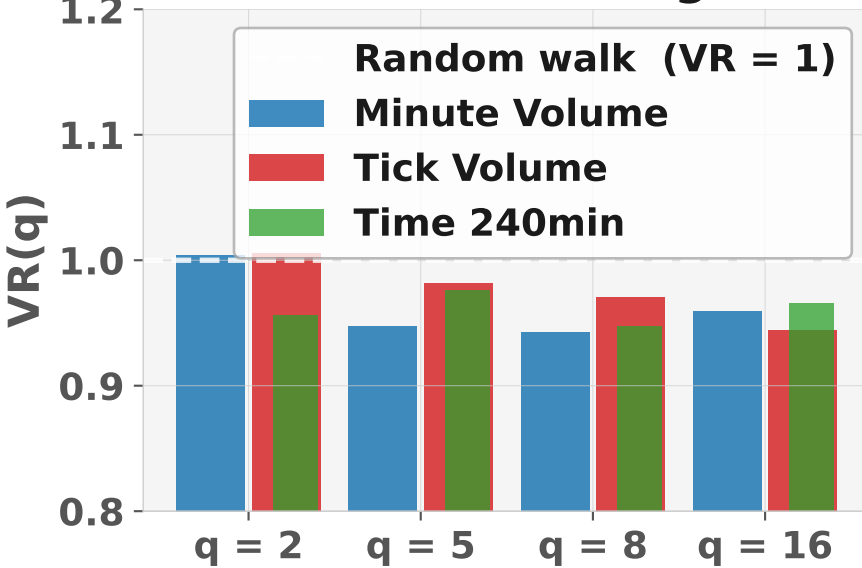
Close Price — A: 5.0 bars/day · B: 10.3 bars/day · C: 6.0 bars/day · CV lower = more uniform · Frequency = more information · Frequent Minute Valuation with market activity = good



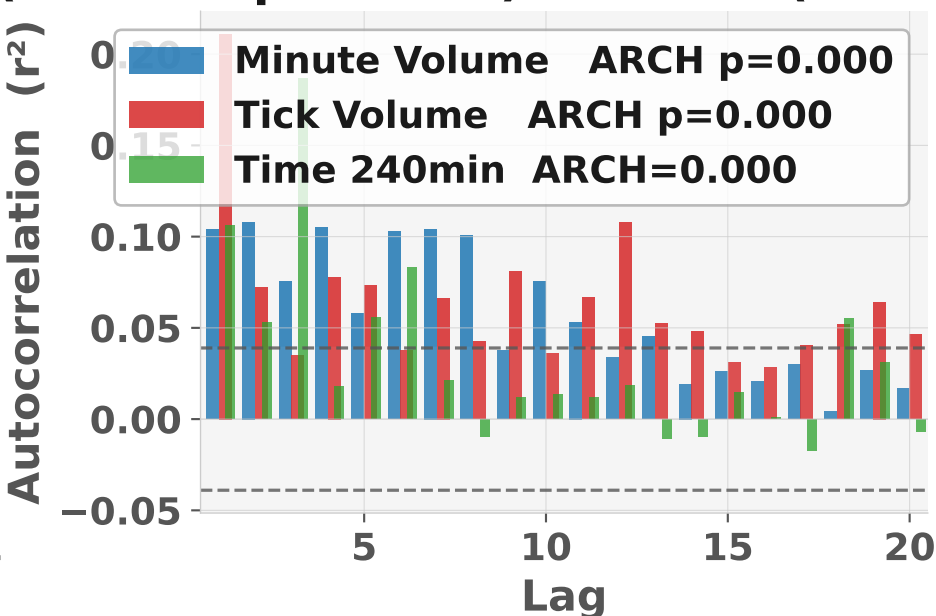
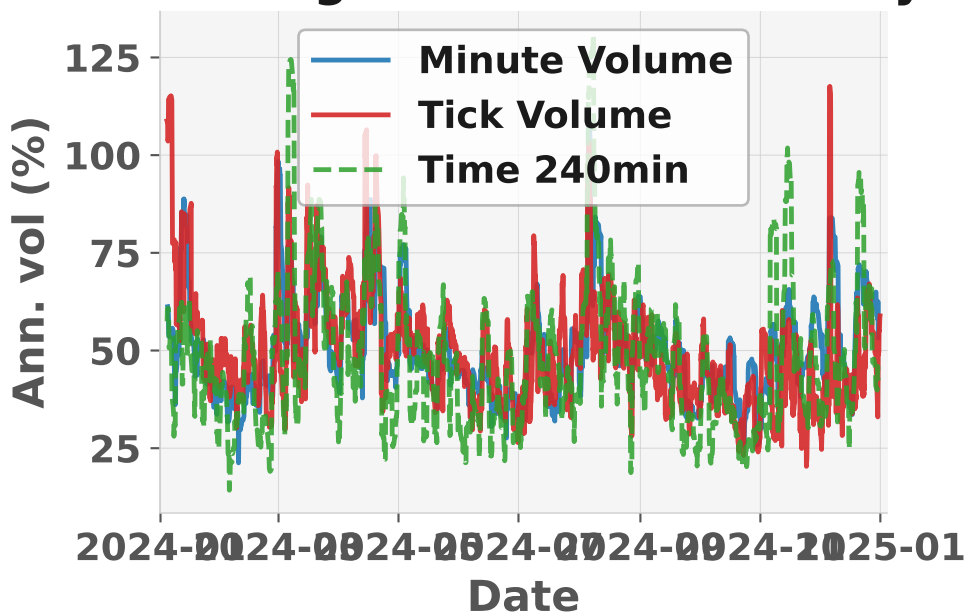
Return Distribution (dashed = fitted normal) ACF (Ljung-Box test) A=0.713 B=0.971 C=0.003



Random walk (VR = 1) VR(q) = |VR(5) - 1| / (VR(5) + 1) A=0.053 B=0.018 C=0.024 · closer to 1 = better



Rolling Annualised Volatility (window 60 days) Return ACF (aligned with Minute Volume leads on 4 of 9 criteria)



Criteria	Minute Volume	Tick Volume	Time
Bar size CV	0.156	0.337	n/a
Entropy	4.582	4.050	3.713
Kurtosis	0.706	1.952	6.025
AC lag-1	0.004	0.005	0.045
LB p (10)	0.873	0.971	0.003
JB stat	38.170	645.690	3321.490
VR(5)-1	0.053	0.018	0.024
Timeout %	4.100	1.800	n/a

Green = leads on this criterion · Yellow = tie
Leading on more criteria $\neq$ better trading performance